

MID TERM EXAMINATION

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 100 | Date: 14-03-2026

Total Questions: 16 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQS

1. Which of the following is NOT listed as an "externally visible" property of a software element according to the definition of software architecture? [2 Marks]
 - a) Provided services
 - b) Performance characteristics
 - c) Internal data structures
 - d) Fault handling

2. What is a primary purpose of architecture documentation in the context of system analysis and construction? [2 Marks]
 - a) To serve as marketing material for potential investors.
 - b) To instruct implementers on what to implement.
 - c) To provide a historical record of architectural trends.
 - d) To predict future software vulnerabilities.

3. In the context of the Software Architecture Business Cycle, what specific skill, beyond technical expertise, is explicitly stated as essential for architects when engaging with stakeholders? [2 Marks]
 - a) Advanced programming language proficiency
 - b) Database administration
 - c) Diplomacy
 - d) Graphic design

4. According to the text, which of the following is a characteristic of a Reference Model? [2 Marks]
 - a) It is a description of element and relation types with constraints.
 - b) It is a division of functionality together with data flow between the pieces.
 - c) It is a mapping of functionality onto a system decomposition.
 - d) It explicitly captures the behavior of each element.

5. Which architectural style is characterized by a centralized data repository accessed by various software components? [2 Marks]
 - a) Layered Architecture Style
 - b) Object-Oriented Architectural Style
 - c) Client-Server Style
 - d) Data-Centric Style

SECTION A: SHORT ANSWERS

6. List three stakeholders' concerns that influence software architectures, as mentioned in the text. [5 Marks]
7. Explain the primary distinction between an "architectural pattern" and an "architecture" itself, based on the provided definition. [5 Marks]
8. Name and briefly describe two types of Module Views. [5 Marks]
9. What are "architectural drivers" in the context of the Attribute-Driven Design (ADD) method? [5 Marks]
10. List three advantages of using an Object-Oriented Architectural Style. [5 Marks]

SECTION A: LONG ANSWERS

11. Define software architecture and elaborate on two key implications of this definition as presented in the text. [10 Marks]
12. Explain the concept of Quality Attribute Scenarios, detailing its six parts and distinguishing between general and concrete scenarios. [10 Marks]
13. Describe the purpose of "Component-and-Connector Views" and list five properties commonly associated with elements in these views. [10 Marks]
14. Explain the importance of documenting software architecture by detailing its three fundamental uses. [10 Marks]
15. Describe the relationship between Architectural Patterns and Tactics, as outlined in the text. Provide an example of how an architectural pattern is determined. [10 Marks]
16. Discuss the four major factors that influence software architectures as described in the "Software Architecture Business Cycle" section. Provide a brief explanation for each factor. [15 Marks]

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